

SUSTAINABILITY ANALYSIS OF ELECTRIC VEHICLES AS OPPOSED TO INTERNAL COMBUSTION ENGINE VEHICLES

James Stanley

School of Mechanical, Industrial,
and Manufacturing Engineering
Oregon State University
Corvallis, OR

Luncheng Zheng

School of Mechanical, Industrial,
and Manufacturing Engineering
Oregon State University
Corvallis, OR

Nomenclature

LCA Life Cycle Assessment
LCI Life Cycle Inventory
LCC Life Cycle Cost
S-LCA Social Life Cycle Assessment
EV Electric Vehicle
ICEV Internal Combustion Engine Vehicle
EoL End of Life
GHG Greenhouse Gas
CO₂ Carbon Dioxide

Introduction

In recent years, there has been a major shift in personal transportation methods, from ICEVs to electric vehicles. Hybrid and electric vehicles now make up over 20% of vehicles sold worldwide [1], on a clearly upward trend for the future [2]. However, EVs differ in some major ways from ICEVs, namely in their power plant. EVs make use of a rechargeable battery pack to power electric motors that drive the wheels [3]. This differs from a typical ICEV that makes use of an engine that burns fossil fuels to provide power.

It is crucial to compare these two transportation platforms, as combined they represent the vast majority of vehicle types used daily. While EVs seem to be constantly proposed as a less environmentally impactful, more sustainable option, there has to be further analysis to prove such statements. Gasoline and diesel fuel

consumption accounts for up to 31% of CO₂ emissions in the US alone [4]. This major sector of environmental impact will continue to evolve and change, hopefully reducing the transportation industry's impact on the environment.

While the impact of internal combustion engines is well known, mainly in fossil fuel extraction and pollutant emissions [5], it is important to note there are some major drawbacks to the electrification of vehicles. Firstly, in the material sourcing and mining that must occur in order to produce batteries. Then there's also the major power draw that constantly charging these batteries presents to the power grid. Lastly, there exists a major infrastructure issue with EVs. The infrastructure exists plentifully for refueling ICEVs, though it is still in development for EVs. This presents transportation problems, especially in more rural areas [6].

Prior research done has found varying results. In some cases LCA has shown environmental benefits of EVs over internal combustion, emitting fewer greenhouse gases, less air acidification, and less fossil fuel consumption. However, EVs showed worse impacts on toxicity and eutrophication from mining practices [7]. Others have found that emissions of EVs in comparison to ICEVs varied from less to equivalent, depending on the major power

grid sources of the region. Areas with coal-dominated electricity showed much less improvement. Overall, showing EV improvements are also tied to the future improvement of manufacturing processes and clean energy in the power grid [8].

Therefore, this project aims to evaluate whether electric vehicles can be considered a truly sustainable alternative to ICEVs by analyzing their environmental, economic, and social performance.

Environmental Analysis Methodology

In order to characterize the environmental performance of EVs and ICEVs, an analysis of prior work was first conducted. What was found was an extensive body of research already conducted in this specific area of study.

These prior studies examined a wide range of vehicle types, functional units, impact methods, and many more variables [9]. From the vast datasets already collected, this project focused on an approach not as often seen, bringing together other analyses into one cohesive paper. The goal is to predict the environmental performance from a wider pane of view, instead of with respect to a specific car in a specific scenario. This is as opposed to conducting and generating new LCA data.

Across the studies analyzed, it was clear that many found that EVs provide benefits, but only after a certain distance of driving [7], [8]. However, ICEVs were often found to perform better in the production phase.

For this environmental analysis, an LCA was conducted to analyze the impact of an EV and ICEV during different life stages. The system boundaries were set to include manufacturing, use, transportation, and EoL. This makes it a cradle-to-grave analysis. This system boundary was chosen so as to include all environmental impacts throughout the lifetime of the vehicle. EVs

and ICEVs outperform each other differently in different life stages [10]. If, for example, manufacturing is not considered, results can vary greatly.

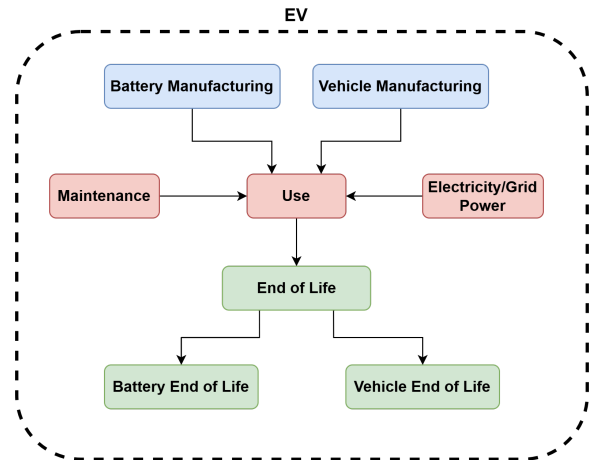


Figure 1: EV system boundary.

The EVs, as shown in Figure 1, have slightly different considerations during the vehicle's life stage. Namely, EVs have the addition of battery manufacturing and EoL, whereas Figure 2 shows that ICEVs do not.

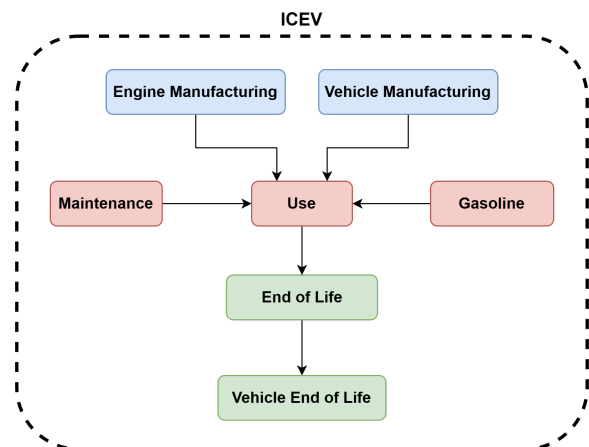


Figure 2: ICEV system boundary.

For the functional unit, a lifetime driving distance was chosen to be 150,000 km. This value is one of the most common standards in vehicle LCAs as the distance before the end of life of a vehicle [3], [5]. However, it is fair to acknowledge that this

is an average distance, and a vehicle's life will depend on many factors. For those studies not matching the set functional unit, data will be normalized.

In order to gather sufficient data for this LCA, 5 independent studies were chosen to be the source of LCI data. Table A1 summarizes the main information regarding the literature referenced for data and final outputs. As well as major assumptions within each article are listed. Table A2 gives the data present in each article used to compare final results.

One assumption incorporated in this LCA was the vehicle type chosen to be analyzed. As the want is to generally give a recommendation on the better vehicle type, not a specific vehicle, a generic mid-sized passenger car was chosen. The assumption is that the vehicle shares the same body for EV and ICEV variants that only differ in powertrain. For ICEVs, the general equivalent is a Euro 5 petrol standard vehicle.

For each analysis, impact methods differed. Some used widely varying ranges, including a large number of categories; others only assessed the climate impact. For this analysis, three impact categories were considered. The first and most important category was the climate impact. As was earlier stated, vehicle emissions comprise such a large amount of CO₂ output that this is one of the biggest areas of environmental impact. This is the only category assessed in all literature found. The second and third categories were human toxicity and eutrophication, respectively. These, while not always reported, are important metrics to quantify the impact of mining and battery manufacturing [1].

Economic Analysis Methodology

In this study, we evaluate the economic feasibility of EVs and ICEVs using an LCC framework. We selected LCC

because it reflects the full financial burden seen by real vehicle owners, rather than just the purchase cost or energy consumption. By considering multiple cost components over the entire period of use, LCC provides a more realistic assessment of long-term affordability.

For consistency, we define the functional unit as 150,000 km, which is a typical lifetime mileage for privately owned vehicles. Based on this functional unit, we examine costs according to the following structure:

$$LCC = C_{\text{purchase}} + C_{\text{energy}} + C_{\text{maintenance}} + C_{\text{battery}}$$

Each term represents a specific aspect of vehicle ownership. Purchase cost includes the initial price and associated taxes. Energy cost refers to the electricity consumption for EVs or the fuel use for ICEVs. Routine maintenance includes scheduled servicing, repair costs, and insurance costs. Repair costs reflect unplanned failures and collision repairs. Insurance costs are determined by long-term premium trends, and battery cost represents the EV-specific economic burden arising from battery manufacturing, degradation, and potential replacement.

This structure allows us to identify not only the areas where EVs have economic advantages, such as reduced routine maintenance, but also the areas where they face disadvantages, including higher repair severity and battery-related depreciation. By analyzing each component independently and then synthesizing them within the LCC framework, we aim to present a comprehensive economic comparison.

Social Analysis Methodology

The S-LCA methodology evaluates social and socioeconomic impacts throughout the vehicle supply chain. This

assessment follows the frameworks described by Molhuizen [27], Gompf et al. [25], and Karlewski et al. [26]. Because EV supply chains rely heavily on minerals such as cobalt, lithium, and nickel, the earlier stages of EV production are associated with substantial social risks. Previous studies have documented unsafe working conditions, toxic exposure, and child labor violations in cobalt mining regions, especially in the Democratic Republic of Congo [26], [27].

S-LCA examines indicators such as worker safety, labor rights, community environmental health, employment creation, and supply chain resilience. As will be shown, EVs and ICEVs differ significantly in these dimensions due to differences in material requirements and industrial structure.

Environmental Analysis Results

The results of the environmental LCA highlighted several key contributions to the impact of each vehicle type. Figure B1 gives a side-by-side comparison of the overall climate change impact of each study reviewed. Every green bar is an EV scenario, and every red bar is an ICEV scenario, as will continue as the custom for this LCA. It can be seen that in studies [7], [9], [10], and [11], the EV is the clear winner when it comes to carbon output over a 150,000 km lifetime. However, study [8] shows that there are some scenarios where the ICEV results in lower overall CO₂ emissions.

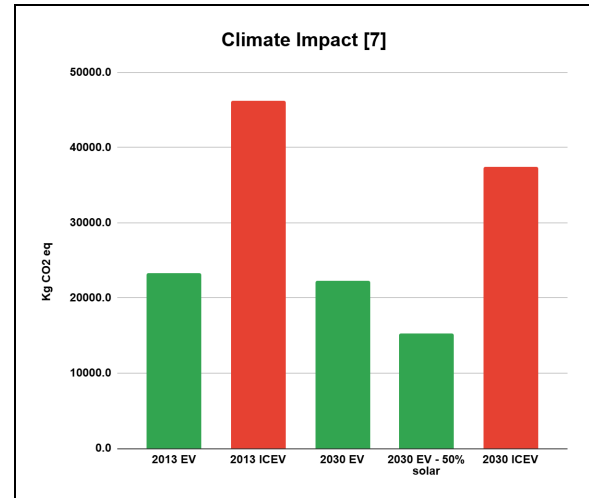


Figure 3: Climate impact results [7]

Study [7] compares a current state scenario in 2013 and a hypothetical scenario in 2030 where grid energy is slightly cleaner and more renewable. Figure 3 shows that EVs overall outperform lifetime CO₂ emissions as compared to ICEVs but both see improvements with cleaner grid energy.

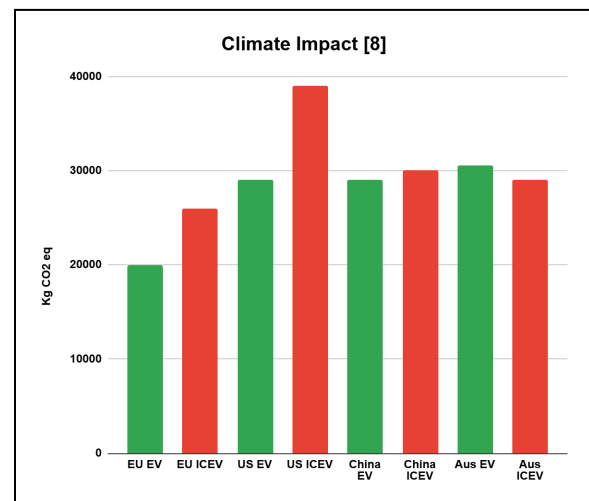


Figure 4: Climate impact results [8]

Study [8] looks at a larger spread of worldwide grid energy types, as shown in Figure 4. The power grids of the EU, US, China, and Australia were used to compare the impact of the same EV and ICEV. Figure 4 highlights the major impact grid energy type can have on a vehicle's lifetime carbon

output. The same EV that overperforms the ICEV in the EU with a more renewable-based grid underperforms in Australia with a more coal-powered grid. This further shows the importance of the grid energy type being used during the life stage of a vehicle.

Study [9] analyzed the impact in Poland and the Czech Republic, both currently and in the theoretical future. It was found that EVs perform better in CO₂ emissions but worse in human toxicity, eutrophication, and particulate matter [12]. Study [10] presents a similar analysis in Europe, focusing on one time frame and resulting in the same conclusions. Study [11] takes a different approach in Brazil, looking at the impact of different driving styles. Based on a light, medium, and aggressive driving style, it was shown that the EV always overperformed in CO₂ but both the EV and ICEV performed worse with more aggressive driving styles.

While the total lifetime CO₂ emissions trend is higher for ICEVs, analysis of each life stage's contribution highlights caveats to this result. As outlined in Figure 5, many of these analyses found the manufacturing phase to contribute much more heavily for an EV as compared to an ICEV. This is largely due to the battery manufacturing involved and is why if a battery replacement is considered, the EV in almost all scenarios will emit a higher CO₂ amount than the ICEV [8].

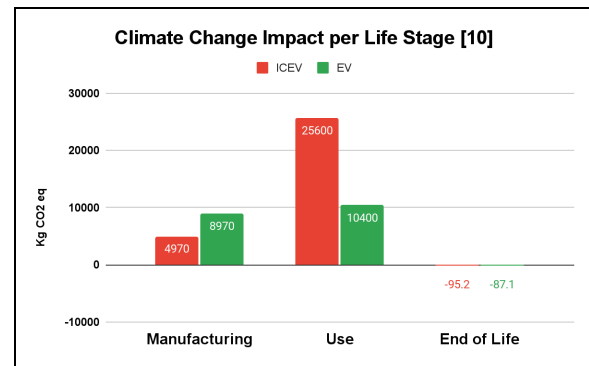


Figure 5: Life stage CO₂ impact [10]

However, the EV shows a much lower use-phase impact compared to the ICEV, which is where the benefits mainly occur. Depending on the electricity grid type, 60,000+ km of driving were needed before the EV was shown to start producing benefits in regard to the CO₂ output [8]. If the EV is driven less than this, no matter the lower impact during the use phase, the overall climate change impact was found to be worse. For end of life, both vehicles were assumed to be recycled and the impact was positive though almost negligible on the scale of the rest of the life cycle impacts.

Figure B2 outlines the human toxicity impacts, which directly oppose those of the climate change impact. In all scenarios, the EV resulted in a higher impact. While each study used different methods of analysis, resulting in the difference in scale seen, the overall trend of the EV having a much worse impact is shown. The majority of this impact comes from the battery manufacturing phase and the sourcing of materials involved in that process [7].

While not as common of an impact category, Figure 6 shows how eutrophication trends the same direction as human toxicity. EVs, even in multiple different power grid scenarios, are shown to consistently perform worse than ICEVs. This further shows how a cradle-to-grave analysis is necessary to see

the full range of impacts during a vehicle's lifetime.

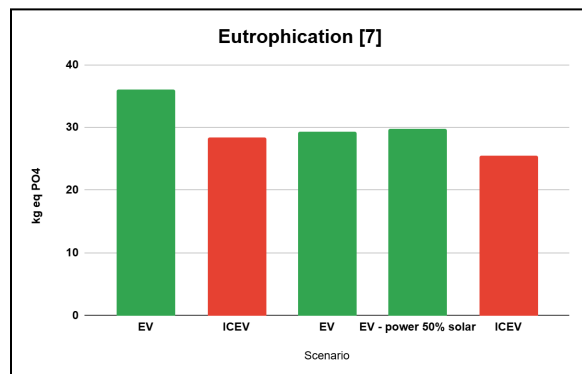


Figure 6: Eutrophication impact [7]

Overall, this data shows that EVs can outperform ICEVs with respect to carbon emissions if:

- The grid power is clean
- The battery is not replaced
- The vehicle is driven for a full 150,000 km lifetime

There are further benefits if the driving is not aggressive and charging is done using solar power. However, ICEVs outperform EVs in human toxicity and eutrophication and are clearly the better option as far as the manufacturing phase is concerned.

From this analysis, it can be seen that there are several areas needing improvement. Battery technology is one area that could alone solve many of the problems that exist. If new, more sustainable materials are transitioned to, many of the negative impacts could be mitigated and EVs made that much better. Another area that needs focus is charging infrastructure. Using solar or at least more renewable sources is paramount to EVs being able to retain their use-phase advantage and outperform ICEVs. Without these changes being explored, ICEVs still remain a competitive option.

Social Analysis Results

The S-LCA comparison reveals that EVs are socially beneficial during the use phase due to reduced urban air pollution. However, they introduce significant social burdens during the manufacturing phase. Table A3 summarizes the findings based on [25]-[27].

EV battery mineral extraction occurs in regions where mining oversight is limited, resulting in significant worker safety and human rights issues [25]. Communities near extraction sites frequently experience environmental degradation due to heavy metal contamination and unsafe waste disposal [26]. Furthermore, EV supply chains are vulnerable to geopolitical disruptions because rare-earth elements are concentrated in a small number of producing countries [27].

ICEV supply chains, while not without social or environmental impacts, generally operate within more stable and regulated industrial sectors. Their global distribution also reduces concentrated social risk, offering greater long-term stability [26], [27].

Economic Analysis Results

Routine Maintenance Costs

We began our economic assessment by reviewing routine maintenance, where EVs demonstrate a well-documented advantage. Data from the U.S. Department of Energy indicates that EVs require around 6.1 cents per mile of scheduled maintenance, compared to 10.1 cents per mile for ICEVs as outlined in Table 1 [13]. When applied to 150,000 km (approximately 93,200 miles), these values correspond to roughly 5,700 USD for EVs and 9,400 USD for ICEVs.

These differences arise from fundamental mechanical distinctions. EVs lack spark plugs, timing belts, oil filters, exhaust systems, and complex lubrication

circuits. As a result, many conventional maintenance procedures simply do not apply to EVs. Additionally, regenerative braking extends the lifespan of the brake system. Consumer Reports supports this finding, showing that EV owners spend approximately half the maintenance costs of gasoline vehicle owners [14].

Vehicle Type	Maintenance (cents/mile)	Total over 150,000 km
EV	6.1	≈ 5,700 USD
ICEV	10.1	≈ 9,400 USD

Table 1: Vehicle maintenance costs

Although this advantage is substantial, routine maintenance is only one part of the cost structure, and the economic picture becomes more complex when we consider repairs, battery costs, and insurance.

Repair Costs

Repair cost represents one of the most significant economic drawbacks for EVs. The Energies study explains that EVs tend to experience fewer mechanical failures, but they incur much higher costs per failure due to their high-voltage architecture and component integration [15]. Even relatively minor collisions can lead to expensive repairs, particularly when they involve the battery casing, inverter assemblies, or thermal management systems.

Industry repair databases offer solid numerical evidence. Mitchell's Q1 2024 report indicates an average EV repair bill of 6,066 USD, compared to approximately 4,700 USD for ICEVs. This makes repair bills approximately 29% higher for EVs [16]. Insurance datasets further reveal that the severity of collision repairs for EVs is often 20-30% higher than that for traditional vehicles [17]. An academic study in *Accident Analysis and Prevention* reports a 6.7% higher per-claim repair cost for EVs

even after accounting for accident type and driver characteristics [18]. This data is further outlined in Table 2.

Indicator	EV	ICEV	Difference
Average collision repair bill	6,066 USD	4,700 USD	EV +29%
Industry repair severity	20–30% higher	baseline	EV more expensive
Academic estimate	+6.7%	baseline	EV more expensive

Table 2: Collision repair costs

The increase in the cost of electric vehicle maintenance mainly manifests in the following aspects: [22]

1. High-voltage safety requirements significantly raise labor costs.
2. EV components (battery, inverter, motor) are integrated and expensive to replace.
3. Limited aftermarket parts reduce opportunities for lower-cost repairs.

These characteristics show that EVs do not simply shift maintenance costs to lower levels, they shift risk toward fewer, but far more expensive, repair events.

Battery Cost

Battery cost remains the dominant economic challenge for EVs. A life cycle cost review reveals that battery production continues to be one of the largest contributors to the EV purchase price, even as technological advancements reduce the cost per kWh [23]. Current estimates suggest that mid-sized battery packs (60-80 kWh) typically cost between 9,000 and 12,000 USD, exerting upward pressure on

the purchase cost (C_{purchase}) and long-term depreciation.

The paper emphasizes how sensitive the EV manufacturing cost is to fluctuations in battery raw materials, strengthening the view that batteries pose both fixed and variable economic risks [23]. Battery degradation also leads to a reduction in resale value, even when replacement is not needed within the 150,000 km functional unit. Moreover, in case of thermal damage or structural impact, full pack replacement becomes essential, driving repair costs to extremely high levels.

For these reasons, the battery is not just a physical component; it brings financial uncertainty, affecting insurance premiums, depreciation, and the perceived long-term risk.

Insurance Costs

Insurance represents another major contributor to LCC. Since insurers base premiums on expected claim severity, the higher repair costs associated with EVs directly increase annual insurance expenses. A 2024 insurance market analysis reports that EV insurance rates increased by 14% year-over-year, with the average repair severity 22% higher than that of ICEV vehicles [24].

Furthermore, the complexity of EV technology, including high-voltage battery systems and advanced electronics, requires specialized repair facilities and technicians [23]. This specialized labor and equipment add to the overall repair costs, which are then passed on to consumers in the form of higher insurance premiums. In addition, the relatively limited availability of replacement parts for EVs, especially in the early stages of their market penetration, can also drive up repair costs and insurance rates. As the EV market matures and more replacement parts become readily available, it is expected that

insurance costs may gradually stabilize or even decrease over time.

The higher vehicle purchase price also contributes to increased replacement cost, further elevating premiums. AAA's *Your Driving Costs* study similarly concludes that although EVs offer low routine maintenance, their insurance and depreciation costs raise their overall cost per mile above that of many ICE vehicles [19].

Therefore, insurance amplifies the economic disadvantages that originate in the repair category.

LCC Case Studies

To validate our overall vehicle type findings, we examined lifecycle cost case studies with specific vehicles as well. The Swedish analysis reviewed found that the Tesla Model Y (EV) incurred a total LCC of 93,100 USD, whereas the Volkswagen T-Roc (ICEV) required only 67,400 USD, making the EV's lifecycle cost 38% higher [21]. These results emerged despite lower EV maintenance and energy cost, demonstrating the impact of purchase price, insurance, and taxation. This data is shown in Table 3.

Vehicle	Purchase (USD)	Operation	Maintenance	Total LCC
EV	59,700	4470	28,800	93,100
ICEV	35,600	14,200	17,700	67,400

Table 3: Total LCC

The research data shows that even though the maintenance cost of electric vehicles is cheaper, the total cost of life cycle is higher than that of traditional internal combustion engine vehicles because of the high maintenance cost and high purchase cost.

Comparative Economic Analysis

Across the combined evidence, a consistent economic pattern emerges. EVs

offer meaningful savings in routine maintenance, but these savings are outweighed by higher repair costs, increased insurance premiums, elevated purchase prices, and the financial uncertainty associated with battery packs. When the cost of battery replacement is included in the LCC framework for trams, EVs frequently exhibit a higher total cost of ownership over the 150,000 km functional unit.

The Swedish case study serves as a demonstration of this effect, illustrating how EVs can remain more expensive even when their operational efficiency is superior. Our results therefore show that economic feasibility must be evaluated holistically, taking into account not just day-to-day savings but also long-term risks and infrequent but high-severity financial events.

Discussion

For the LCA, the approach of summarizing data already collected is not as common, though for a topic so heavily researched, it proved to be very beneficial. Instead of this analysis providing results on a certain car driving in a certain part of the world, the results were able to comment to a greater degree on EVs and ICEVs in general. This fed into the goal of providing a consumer an overall knowledge of vehicle types as a whole and which to consider.

While this study showed some clear trends, future work would benefit heavily from incorporating even more datasets. More data would further reinforce the results of these findings and possibly point out more problem areas needing analysis.

However, at the same time, the variability of data was one of the bigger limiting factors of this analysis. It can be difficult to compare one study to another if they use different functional units or especially different impact methods and datasets.

Though the time commitment would be heavy, a step forward could be an LCA conducted from the ground up incorporating all life phases, multiple power grids, and impact methods. Along with incorporating several different categories of assessment to look at more areas than climate change, human toxicity, and eutrophication.

The economic analysis provided a strong example of the vehicle's costs both by looking at overall average costs and specific vehicle cases. By diving into each cost contributor, the whole picture could be seen. Without such a breakdown, EVs might not seem so clearly more expensive. However, it is important to state that costs are highly dependent on many variables and these are averages. Repair costs for one user are almost always different from another. As well as insurance depends heavily on the user's driving history and other factors.

An analysis using many data points on specific vehicles that was then synthesized into an overall economic impact could help this issue. Such a study could not only give overall trends but point out which vehicles specifically result in higher or lower costs. This could then help inform consumers on the specific vehicles to avoid resulting in a better economic outcome.

The social analysis showed important outcomes of the EV and ICEV life stage. However, the largest issue is the qualitiveness of such an analysis. Finding ways to bring in quantitative values to better “score” each vehicle's social performance may give more sway to the findings.

Another interesting advancement would be the incorporation of interviews or surveys into the social analysis. Getting a large net of consumer opinions could help in telling trends of social impact or finding areas missed. This would however be a large undertaking to gain good data from many sources.

Conclusions

Overall, this analysis has shown that EVs can provide some environmental benefits though their battery architecture prevents them from truly outperforming ICEVs. While the use-phase impact is considerably less for EVs, the driving distance and aggressiveness are not easy to quantify. Whereas the negative impact from making the car is a given no matter what. While there is hope for EVs presenting a better environmental option eventually, currently it cannot be said that they are better than ICEVs.

For economics, based on our analysis, we interpret EV performance as a balance between low daily cost and high overall cost. Routine maintenance and energy efficiency work in favor of EVs, but these benefits are overshadowed by the combined effect of expensive battery systems, high repair severity, and elevated insurance premiums. In particular, the possibility of large, infrequent expenses makes EVs financially riskier than ICEVs from an LCC perspective.

Thus, under present market conditions and in the absence of substantial government subsidies, EVs do not surpass ICEVs in overall economic feasibility over a 150,000 km lifetime.

The S-LCA results show a distinct trade-off in social sustainability. EVs offer clear downstream benefits through reduced urban air pollution, yet impose large upstream social burdens linked to mineral extraction, including risks to worker safety, child labor, environmental degradation in mining regions, and increased geopolitical vulnerability. ICEVs, while environmentally less favorable, exhibit more stable and regulated supply chains with lower concentrated social risks.

Overall, this study shows that from an environmental, economic, and social perspective, EVs do not offer such

substantial benefits as to entirely replace ICEVs. There exist many issues that make ICEVs and hybrid vehicles still a highly competitive option. Without further advancement in battery technology, a push for renewable energy, reduction in vehicle cost, and tightening of mining regulation processes, EVs still have a long way to go.

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Appendix A - Tables

Table A1

Reference	System Boundary	Functional Unit	Vehicle	Impact Methods	LCA Tool	Assumptions
[7]	Cradle-to-Grave	150,000 km	Lower-medium passenger car Same body ICEV/EV 24 kWh Battery Euro 5 gas	IPCC/CML 2001	Ecoinvent	- Same body for ICEV and EV - Excludes battery recycling - Considered road impacts - Battery will not need replacement - Fueling/charging infrastructure excluded
[8]	Cradle-to-Grave	0 - 200,000 km	Generic reference vehicles 35.8 kWh battery C-segment Same body ICEV/EV	CO2 Impact Only	Spreadsheets /GaBi	- Assumed regional electricity mix - Same body for ICEV and EV - Maintenance intervals are averages
[9]	Varies (Cradle-to-Grave used only)	1 km	Varies	Varies	Varies	- Varies
[10]	Cradle-to-Grave	150,000 km	Gasoline turbocharged 1.5L EURO 5 Li-ion battery - 85 kW motor	ILCD	GaBi 6.3	- Assumed regional electricity mix - EoL of Batteries ignored
[11]	Cradle-to-Grave	150,000 km	2022 Nissan Leaf 40 kWh battery Chevrolet Cruze Sport6 1.4T	IPCC (GHG-100)	REET/Ecoinvent	- Three average driving styles - Average electricity mixes - Vehicle range

Table A2

Scenario	Climate Change (kg CO2 eq)	Human Toxicity (kg 1.4-DCB)	Eutrophication (kg eq. PO4)
[7]			
2013 EV	23264.5	20538.6	35.989
2013 ICEV	46209.0	14219.1	28.4
2030 EV	22322.8	19691.4	29.3
2030 EV - 50% solar	15303.4	19495.6	29.8
2030 ICEV	37426.3	12863.5	25.4
[8]			
EU EV	20000	-	-
EU ICEV	26000	-	-
US EV	29000	-	-
US ICEV	39000	-	-
China EV	29000	-	-
China ICEV	30000	-	-
Aus EV	30500	-	-
Aus ICEV	29000	-	-
[9]			
EV 2015	41400	49650	-
EV 2050	25800	35100	-
ICEV	42600	12750	-
[10]			
EV	19300	4050	-
ICEV	30500	85.5	-
[11]			
Eco driving EV	3795.00	-	-
Eco driving ICEV	30761.81	-	-
Standard driving EV	5190.00	-	-
Standard driving ICEV	34121.81	-	-
Aggressive driving EV	6135.00	-	-
Aggressive driving ICEV	45356.81	-	-

Table A3

Social Indicator	EV Supply Chain	ICEV Supply Chain	Reference
Worker Safety & Health	High risk (mining accidents, toxic exposure)	Moderate	[9]
Labor Rights / Child Labor	High risk (DRC cobalt mining)	Low	[9], [11]
Community Environmental Health	High burden (soil/water contamination)	Moderate	[11]
Employment Creation	High-skill job creation	Broad industrial employment	[9]
Supply Chain Stability	Low stability (rare-earth dependency)	High	[10], [11]

Appendix B - Figures

Figure B1

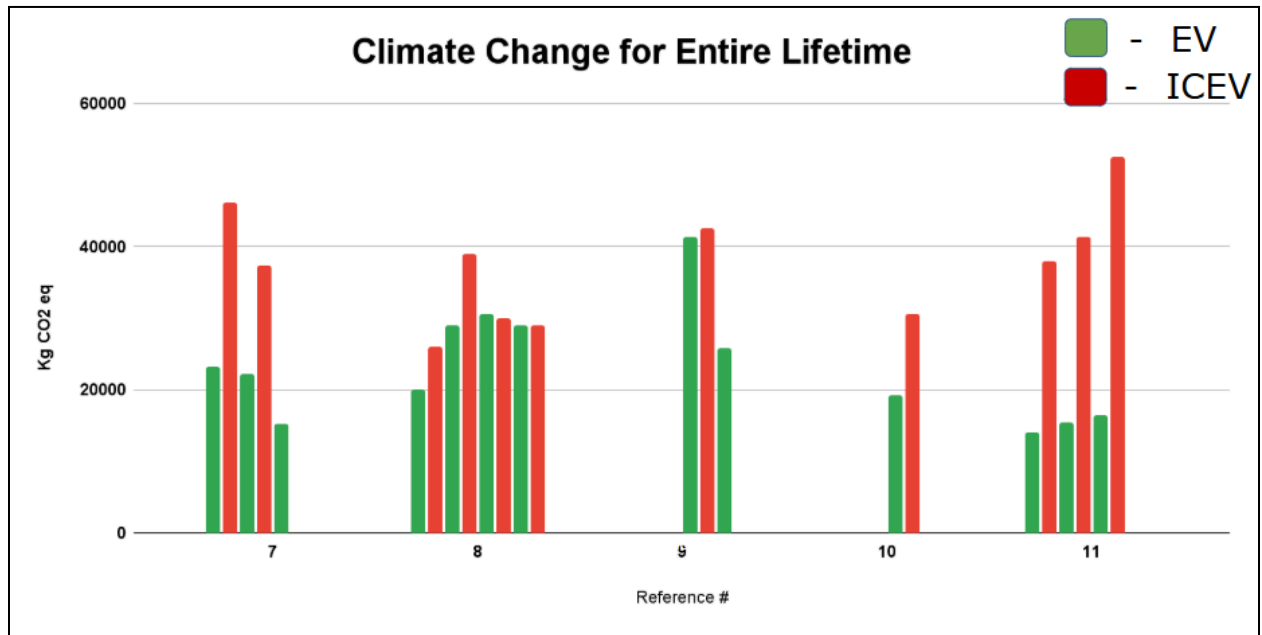


Figure B2

